Lecture 8. Discrete Random Variables

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Families of distributions

- While working out distributions for random variables, you may notice some patterns that you use again and again. A set of distributions that share the same pattern is called *a family of distributions*.
- Mathematicians have given names to these families of distributions and derived formulas whenever possible.

We will study some of the most commonly-used families of distributions here.

The Geometric family

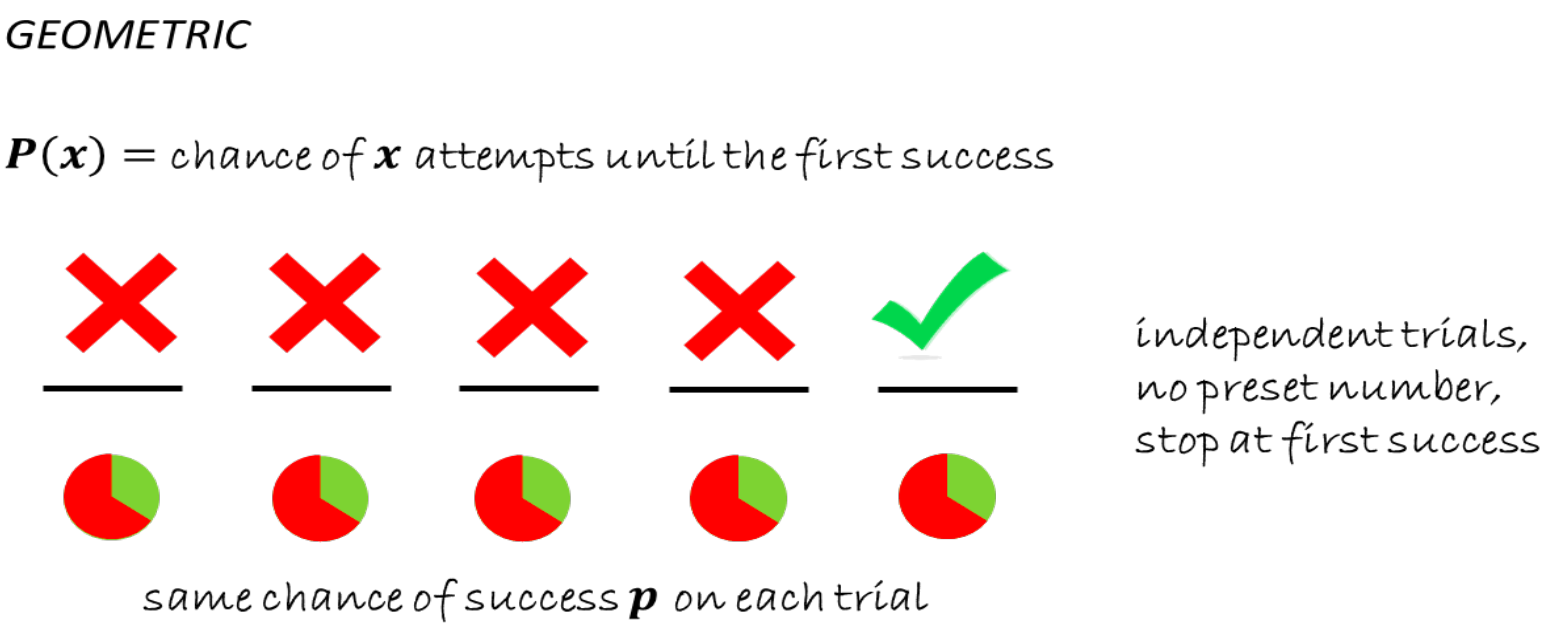
- You may have noticed this pattern when the random variable equals the number of times you repeat an action until something happens.
- Here are some examples you have encountered:
  - the number of times you will toss a coin until you get heads
  - the number of times you will roll a die until you get a five
  - number of files you will search until a keyword is found

The Geometric family

- Details:
  - We ask about the number of times we repeat an action until the result we are waiting for happens.
  - We know the chance of the result we are waiting for, and use the Complement Rule for the chance it does not happen. These are the same on each try.
  - The possible outcomes (number of tries) are 1, 2, 3, 4, and so on (the positive integers).
  - Each try is (or is assumed) independent of the others, and we use the Multiplication Rule.
  - For the probability of each possible outcome, we multiply the chance of the opposite several times, and the chance of the desired result once, at the end.

The Geometric distribution

The following diagram illustrates a Geometric random variable. Note that it shows a numerical outcome of 5 attempts (until the first success).



Example

- You will roll a die until you get a 1 or 2. Find the distribution of the number of times you will roll the die.

- To write down formulas, here are the letters most often used:
  - p= probability of what we are waiting for (when it happens, we stop)
  - q=1−p= probability of the opposite result (when it happens, we do not stop yet)
  - Note that $0<p<1$ and $0<q<1$, because each of these is a probability of something uncertain.

The Geometric distribution

- A distribution that follows this pattern or template is called a Geometric distribution, because the probabilities form a geometric sequence, and when we add them up and then form a geometric series.
- In calculus, you learned a geometric sequence is formed by multiplying the same factor over and over again.
- The factor multiplied over and over again is called the “common ratio.”
- If you choose a “starting term” and want to add up all the entries from the starting term on, use the formula derived in calculus:

$$\text{sum of terms in a geometric sequence} = \frac{\text{starting term}}{1-\text{common ratio}}$$

- With this background information, we can verify that the template above is actually a valid probability distribution.

The Geometric distribution

- We can write down a nice formula for the probability mass function of the Geometric template:

$$P(x) = q^{x-1}p$$

for x= 1, 2, 3 ... and $q = 1 - p$.

- Anytime a random variable's distribution follows the Geometric pattern or template, we can describe its distribution simply by saying it is Geometric, and specifying what value of p to use.
- With that information, everything else is known: the value of q, the probability mass function, and the cumulative distribution function.

Example

For each random variable below, decide if it has a Geometric distribution. If it does, specify p. If it does not, mention why.

(a) The number of times you will roll a die, if you roll until you get a five.

(b) The number of files checked, if a file is checked until a keyword is found, and the chance of finding the keyword in a file is 25%, independently of other files.

(c) The number of times you will draw, if you draw a marble from a jar (and do not put it back) until you get a green one, and the jar has 4 green and 8 blue marbles.

Geometric average and standard deviation

- If a random variable is Geometric,
  - Geometric average outcome =$1/p$
  - Geometric standard deviation = $\sqrt{1-p}/p$

- What happens to the expected value with a higher success rate p? Why?

Geometric average and standard deviation

(a) Find the expected value and standard deviation of the number of tosses of a coin until the first head.

(b) Find the expected value and standard deviation of the number of rolls of two dice until the first doubles.

Example

An employer is interviewing potential candidates for a vacant position and wants to know how many interviews he/she has to conduct until the perfect candidate for the job is found. Let p, the probability that he succeeds in finding such a person, equal 0.20.

1. Let X denote the number of people he selects until he finds his first success. What is the probability mass function of X?

2. What is the probability that the 2nd candidate is perfect for the position?

3. What is the probability that the employer has to interview at least 3 candidates?

The Binomial family

- You may have noticed this pattern when the random variable equals the number of times something happens when there are a fixed number of opportunities for it to happen. The outcome we are interested in is the number of times it actually happens.

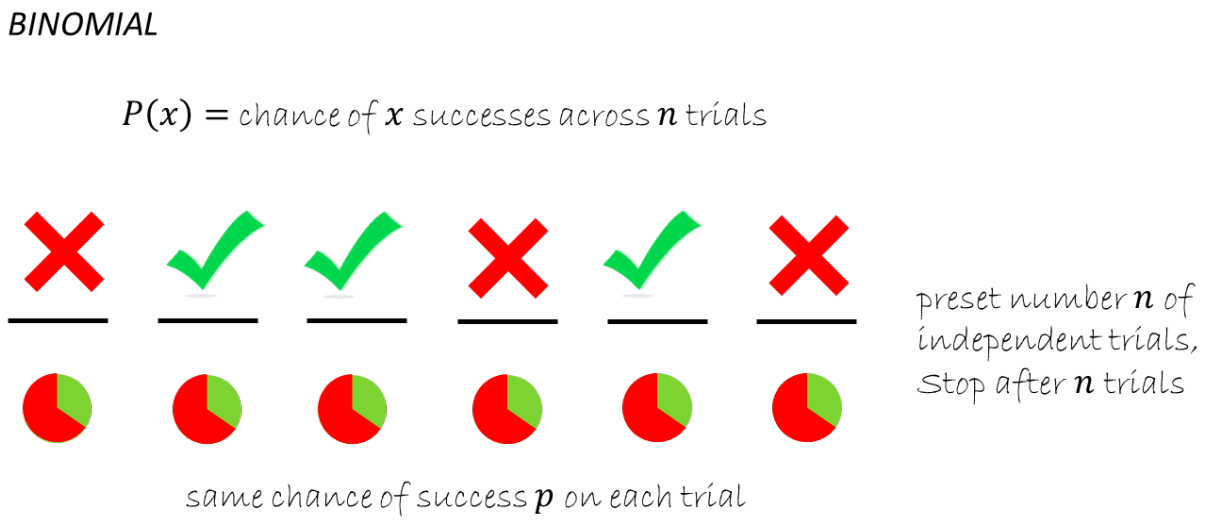
- Here are some examples you have encountered.
  - the number of heads that will occur when a coin is tossed 3 times
  - the number of corrupted bits there will be in a packet of 4

The Binomial Family

- Details:
  - We ask about the number of times something happens. For convenience, refer to the result we are counting as a "success."
  - There is some fixed number of tries, attempts, or opportunities for a success to happen. For convenience, let us say there are n opportunities.
  - We know the chance of a success happening, and use the Complement Rule for the chance of a success not happening. These chances are the same in each try.
  - The possible outcomes (number of times the success happens) are 0, 1, 2, 3, up to n, the maximum number of successes there could be.
  - Each try is (or is assumed) independent of the others, and we use the Multiplication Rule.
  - To compute the probability of each possible outcome, we use both the Multiplication Rule and the Addition Rule, because we have to account for all the arrangements of successes with non-successes.

The Binomial Family

- The following diagram illustrates a Binomial random variable. In the diagram, there turned out to be 3 successes (among 6 trials).



- A distribution that follows this pattern or template is called a *Binomial distribution*, because of the binary nature of each attempt (success, or not) and because of its formula.

Example

We will roll a die 5 times, and we are interested in how many times it will be a "1" or a "2" .

(a) How does this experiment fit the patterns above?

(b) Suppose we want the probability that we will get a "1" or a "2" exactly twice. One way is if we are successful on the first roll and the fourth roll (and not on any other roll). What is the probability this happens?

(c) Another way we could be successful exactly twice is if it happens on the third roll and the fifth roll (and not on any other roll). What is the probability this happens?

Example

(a) If 8 dice are rolled, how many ways could there be exactly 3 fives?

(b) If there are 12 trials, how many ways are there to have exactly 4 successes?

Example

(a) Toss a coin 5 times. What are the chances it will be heads exactly 2 times?

(b) Roll a die 8 times. What are the chances it will be a 1 or a 2 exactly 3 times?

(c) A packet has 4 bits. Each bit has a 10% chance of being corrupted, independently of other bits. What is the probability exactly 1 bit is corrupted?

The Binomial Family

Anytime a random variable's distribution follows the Binomial pattern or template, we can describe its distribution simply by saying it is Binomial, specifying the number of "trials" n, and specifying what value of p to use. With that information, everything else is known, including the value of q and the probability mass function.

Example

Check whether each situation involves a Binomial random variable. Specify n and p.

(a) Roll a die 5 times. We are interested in the number of "1".

(b)A program consists of 8 modules; each module contains an error with probability 0.15 independently of the other modules. We are interested in the number of error-free modules.

(c)A jar contains 4 green and 8 blue marbles. We draw three marbles at random without replacement. We are interested in the number of blue marbles.

The Poisson Family

- This family of distributions is named after the French mathematician Siméon-Denis Poisson (1781-1840), who discovered it by taking the following limit of the Binomial probability mass function.

$$\lim_{n\to\infty} \frac{n!}{k!\,(n-k)!} p^k (1-p)^{n-k} = e^{-\lambda} \frac{\lambda^k}{k!}$$

- We assume that $n \to \infty$ and $p \to 0$ in a way such that $np = \lambda$ constantly throughout the limiting process.
- The resulting function is the probability mass function formula for a Poisson distribution:

$$P(x) = e^{-\lambda} \frac{\lambda^x}{x!}$$

The Poisson Family

- The Greek letter λ (lambda) is used for the variable that controls how this distribution works. We can make it any positive real number we want, but the Poisson distribution best fits our situation if we make λ equal the average outcome (or, the expected outcome). The average outcome will depend on what length of time is involved.

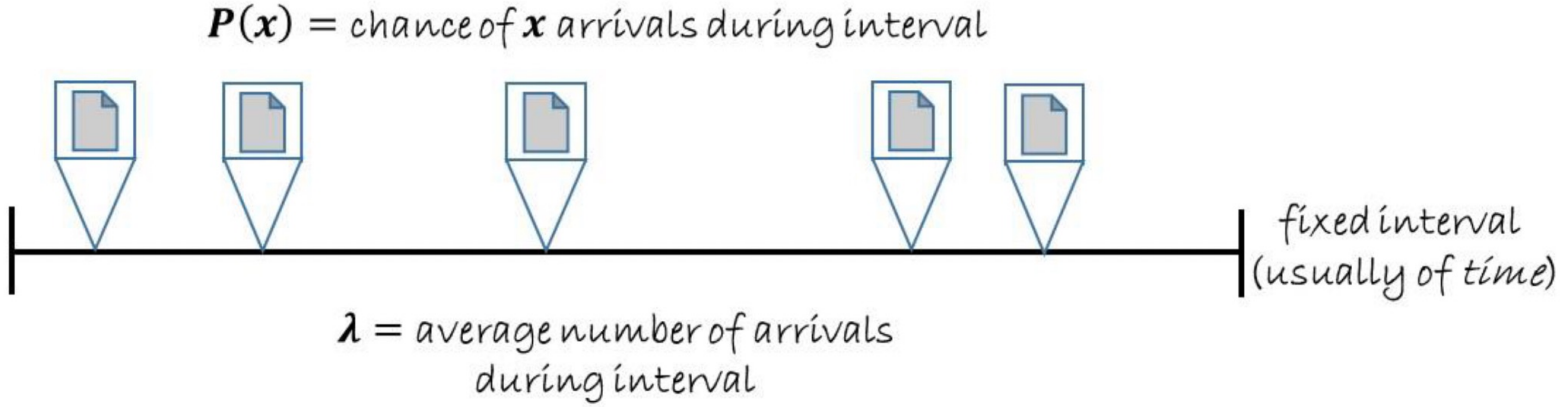
- We use a Poisson distribution for the number of "successes," as we do with a Binomial distribution. But instead of separate, independent trials, the context is a smooth interval of time.

Example

Someone is repeatedly tossing a coin and gets heads, on average, 8 times during any interval of 30 seconds. Use a Poisson distribution to estimate the probability they will get exactly 5 heads during any 30-second interval.

Poisson

- We can use the Poisson distribution to estimate probabilities in a variety of counting situations, so long as we know the average outcome, and more than one "success" cannot occur simultaneously.
  - A typical situation is to estimate chances about the number of arrivals of messages to an inbox,
  - The number of jobs a processor receives, or the number of calls a customer center receives.
- The following diagram illustrates a Poisson random variable. Note that it shows an outcome of 5 arrivals.



Example

During any minute, the average number of jobs a processor receives is 2.5. Use a Poisson distribution for the number of jobs the processor receives.

(a) What is the probability the processor receives 2 or fewer jobs in the next minute?

(b) What is the probability the processor receives 4 or fewer jobs in the next two minutes?